

1. Administrative & Study Identification

Full Study Title: Treatment Landscape and Real-World Dosing Patterns With CDK4/6 Inhibitors for Breast Cancer in Portugal
Short Title/Acronym: PORTRAIT **Protocol Number:** [Not Available/Novartis Internal] **NCT Number:** NCT07223476 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Cyclin-dependent kinase 4/6 inhibitors (CDKI) - Ribociclib, Palbociclib, and Abemaciclib
Phase of Development: Observational/Retrospective (Phase N/A)
Medical Monitor: Study Director, Novartis Pharmaceuticals

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess and characterize the real-world cyclin-dependent kinase 4/6 inhibitor (CDKI) use in advanced/metastatic breast cancer (a/mBC), evaluate treatment persistence, and quantify CDKI dose adjustments in Portugal.
Secondary Objectives: To evaluate the Time to Next Treatment (TTNT) and explore dose reduction patterns across different treatment intervals (0-6 months, 7-12 months, and ≥12 months).

2.2 Background & Rationale To address the need for real-world evidence regarding the treatment landscape of CDKI therapy in HR+/HER2- a/mBC in Portugal. While CDK4/6 inhibitors combined with endocrine therapy are the standard of care, there is a gap in understanding real-world treatment persistence and dosing adjustment patterns outside of controlled clinical trials. This study leverages IQVIA hospital drug consumption data to quantify these patterns.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective, non-interventional cohort study) **Control Type:** Single-arm (Real-world data cohort) **Blinding:** Open-label (N/A for retrospective analysis) **Randomization:** N/A (Non-Probability Sample)

3.2 Planned Enrollment & Duration Target \$N\$: Eligible patients with CDKI consumption across 16 public hospitals **Number of Sites:** 16 public hospitals in Portugal **Estimated Study Start:** 29-Mar-2024 (Data extraction covering Nov 2018 to May 2023)
Estimated Primary Completion: 01-Jul-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients of all ages. 2. Patients with at least one CDKI consumption from November 2018 to May 2023. 3. Patients with consumptions in Gynecology and Oncology specialties. 4. HR+/HER2- a/mBC patients with at least 1 month of ribociclib, palbociclib, or abemaciclib treatment.

4.2 Exclusion Criteria 1. Consumption patterns of target drugs that are not compatible with any known a/mBC treatment regimen. 2. Patients in early stages of breast cancer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Real-world standard dosing of CDK4/6 inhibitors. Patients were categorized as having 0, 1, or 2 dose reductions. **Route of Administration:** Oral **Duration of Treatment:** Up to approximately 4 years and 6 months (based on the data availability window of Nov 2018 to May 2023).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant endocrine therapy (ET) appropriate for HR+/HER2- a/mBC treatment. **Prohibited:** Treatments indicating consumption patterns incompatible with known a/mBC regimens.

6. Study Timeline & Visit Schedule

(Note: As this is a retrospective database study, the following represents the temporal data extraction intervals rather than physical visits). | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Nov 2018 - May 2023 | IQVIA database filtering for Gynecology/Oncology CDKI consumption | | **Baseline** | Day 0 | Index date: First recorded consumption of a CDKI treatment line | | **Interim** | Months 0-6, 7-12, >12 | Categorization of dose reductions (0, 1, or 2 reductions) | | **End of Study** | Up to 4.5 Years | Calculation of Median Treatment Duration and Time to Next Treatment (TTNT) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: - Number of Patients Treated With a CDKI Between November 2018 and May 2023. - Median Treatment Duration. - Median Time to Next Treatment (TTNT) (calculated considering the time between the start of one treatment line and the start of the next treatment line). - Number of Patients With Dose Reductions by Treatment Length per Line of Treatment (0-6 months, 7-12 months, and ≥ 12 months). **Measurement Method:** Retrospective calculation using secondary hospital drug consumption data (IQVIA database).

7.2 Secondary & Exploratory Endpoints - Detailed characterization of CDKI sequence and persistence across multiple lines of therapy.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A for standard AE reporting. Dose reductions/interruptions extracted from the database act as a real-world proxy for tolerability and toxicity. **Serious Adverse Events (SAEs):** N/A (Retrospective database study). **Data Monitoring Committee (DMC):** Supervised by Novartis Pharmaceuticals internal review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible HR+/HER2- a/mBC patients identified in the database who received at least 1 month of a CDKI. **Sample Size Justification:** No formal power calculations; descriptive analysis based on the available real-world IQVIA consumption data from 16 public hospitals. **Handling of Missing Data:** Incomplete or incompatible consumption patterns were excluded from the analysis cohort.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local real-world data regulations. **Monitoring Plan:** Retrospective data validation and quality checks on IQVIA hospital consumption datasets. **Audit Trail:** IQVIA database standard data validation parameters.

11. Study Identifiers & Governance

Primary ID: NCT07223476 **Registry Number:** NCT07223476 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** PORTRAIT **Official Regulatory Title:** Treatment Landscape and Real-World Dosing Patterns With CDK4/6 Inhibitors for Breast Cancer in Portugal

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Advanced/Metastatic Breast Cancer (a/mBC) **Diagnosis Threshold:** Confirmed HR+/HER2- a/mBC receiving treatment **Medical Methodology:** CDK4/6 inhibitors + Endocrine Therapy **Optimization Target:** Characterizing real-world treatment persistence and dose adjustments

13. Study Procedures & Methodology

Management Pathway: Standard a/mBC clinical pathway in Portugal
Education Protocol (HCP): N/A (Retrospective study) **Education Protocol (Patient):** N/A (Retrospective study) **Quality Audit Mechanism:** Algorithmic filtering of hospital consumption data to ensure compatible treatment regimens

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 4 years and 6 months **Onsite Visit Cadence:** N/A (Retrospective data extraction) **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** Data stratified continuously and evaluated at 0-6, 7-12, and ≥ 12 month intervals

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				